



Bhavan's

(Founded in 1978 by Kulapati Dr. K. M. Murali with the blessing of Mohanrao Gadgil)

आ नो भद्राः क्रतावो यन्तु विश्वतः।

Let noble thoughts come to us from every side

Tel : 91-22- 2670 8520

2670 7440

2628 7250

Fax : 91-22- 2670 1422

SARDAR PATEL INSTITUTE OF TECHNOLOGY

Autonomous Institute affiliated to University of Mumbai

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai - 400 058, India

E mail: principal@spit.ac.in Website: www.spit.ac.in

Date: 21/10/2024

To Whom It May Concern:

I am pleased to write this letter of recommendation for Mr. Kevin Doshi, who has been an exemplary student in both academic and research settings. I have taught Kevin the course "Data Structures and Algorithms" in which he earned an A+, reflecting his strong aptitude and passion for the subject. Currently, I am mentoring him on his major project, "WaveSplit," a cutting-edge initiative focusing on audio signal processing using Convolutional Neural Networks (CNNs) and wavelet transformations. Since I have mentored Kevin closely in class and for his project, I am able to comment on his abilities and traits that make him an ideal student for graduate studies. Through this LOR, I would like to explain Kevin's abilities in the context of the project he is currently pursuing.

Kevin's project is one of the few projects that is interdisciplinary in that it combines the use of CNNs and wavelet transformations to extract specific audio segments. The novelty of the approach makes this project noteworthy. Though existing research was sparse, Kevin conducted thorough literature review on the content available to research the applications of wavelet transformations in audio signal processing, comparing traditional Fourier based techniques to his proposed CNN-wavelet hybrid approach. During the project, Kevin showed proficiency in complex algorithm design and implementation, especially in the areas of pattern recognition and feature extraction. He used Python, TensorFlow and wavelet transformation libraries like PyWavelets to create a sophisticated model that optimally balances computational efficiency with signal integrity. In the early stages of the project, his approach ran into performance bottlenecks. Instead of giving up or switching the project, he painstakingly identified ways to improve the integration between wavelet transformed signals and CNN layers allowing the model to maintain high accuracy without compromising the computational speed.

I am happy to report that Kevin has now made significant progress in his project. His ability to overcome challenges and apply Machine Learning concepts to real world audio data while implementing this novel project demonstrates both the depth of his technical expertise and his courage to explore innovative approaches. Additionally, his persistence when faced with challenges and willingness to patiently optimize the approach he believes in are traits that will serve him well in graduate school and beyond.

Overall, Kevin's initiative, drive and commitment to develop effective models through innovative approaches make him an outstanding candidate for graduate studies. I strongly recommend Kevin to your program and am confident that he will make significant contributions to your universities academic and research endeavours.

AA Godbole

Professor Anand Godbole

Department of Computer Engineering

Bhartiya Vidya Bhavans Sardar Patel Institute of Technology

+91 99679 69420 | anand.godbole@spit.ac.in